

Introduction

Sample-size planning for diagnostic studies typically targets **precision** (CI width) for sensitivity and specificity rather than power against a hypothesis. Separate calculations apply to diseased and non-diseased cohorts.

Prerequisites

Sensitivity, specificity, proportion CIs.

Theory

For a Wilson 95 % CI on sensitivity of anticipated value p_{sens} with desired half-width w :

$$n_{\text{dis}} \approx \frac{z_{0.975}^2 p_{\text{sens}} (1 - p_{\text{sens}})}{w^2}.$$

Analogously for specificity using non-diseased. Total sample = $n_{\text{dis}} + n_{\text{non-dis}}$, with the ratio determined by the prevalence of disease in the source population.

For hypothesis-testing framing (is sensitivity > some reference), the one-proportion power formula applies.

Assumptions

- Anticipated sensitivity / specificity from pilot or literature.
- Independent gold-standard verification for every subject.

R Implementation

```
# Design 95% CI half-width 5% on sensitivity = 0.90
p_sens <- 0.90; w <- 0.05
n_dis <- qnorm(0.975)^2 * p_sens * (1 - p_sens) / w^2
ceiling(n_dis)

# Spec = 0.85, same width
p_spec <- 0.85
n_non <- qnorm(0.975)^2 * p_spec * (1 - p_spec) / w^2
ceiling(n_non)

# Hypothesis-test framing: sensitivity > 0.80
# Expected sens = 0.90, alpha = 0.05, power = 0.80
library(MKmisc)
power.prop1.test(p0 = 0.80, p1 = 0.90, sig.level = 0.05, power = 0.80)
```

Output & Results

For sens 0.90 +/- 5 %: 138 diseased cases. For spec 0.85 +/- 5 %: 196 non-diseased. Hypothesis-test $n \approx 130$ diseased cases to show sensitivity > 0.80.

Interpretation

“To estimate sensitivity of 0.90 with a 95 % CI half-width of 5 %, at least 138 diseased cases are required. Assuming a disease prevalence of 30 % in the source population, a total enrolment of 460 subjects provides adequate non-diseased controls.”

Practical Tips

- Plan separately for sensitivity and specificity; the two calculations have independent sample-size requirements per cohort.
- Use Wilson or exact intervals for the reported CI; Wald intervals can be optimistic.
- STARD reporting guidelines recommend reporting both prevalence and the numbers of diseased / non-diseased subjects.
- Paired designs (both tests on each subject) use McNemar-style within-subject comparisons; they are more efficient than unpaired designs.
- AUC-based sample sizes exist (e.g., Hanley-McNeil); different from sensitivity-only calculations.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Sample size, power, and Quarto reporting
- Power: closed-form and simulation